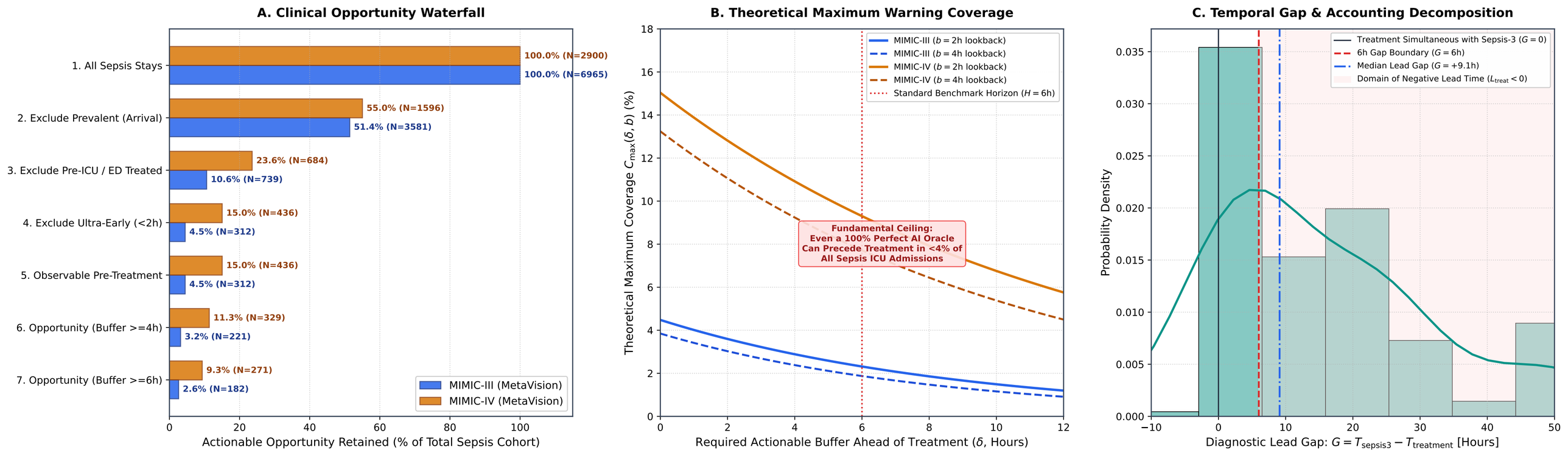
A. Clinical Opportunity Waterfall
1. All Sepsis Stays
2. Exclude Prevalent (Arrival)
3. Exclude Pre-ICU / ED Treated
4. Exclude Ultra-Early (<2h)
5. Observable Pre-Treatment
6. Opportunity (Buffer >=4h)
7. Opportunity (Buffer >=6h)
100.0% (N=2900)
100.0% (N=6965)
55.0% (N=1596)
51.4% (N=3581)
23.6% (N=684)
10.6% (N=739)
15.0% (N=436)
4.5% (N=312)
15.0% (N=436)
4.5% (N=312)
11.3% (N=329)
3.2% (N=221)
9.3% (N=271)
2.6% (N=182)
MIMIC-III (MetaVision)
MIMIC-IV (MetaVision)
0
20
40
60
80
100
Actionable Opportunity Retained (% of Total Sepsis Cohort)
B. Theoretical Maximum Warning Coverage
MIMIC-III ($b = 2$h lookback)
MIMIC-III ($b = 4$h lookback)
MIMIC-IV ($b = 2$h lookback)
MIMIC-IV ($b = 4$h lookback)
Standard Benchmark Horizon ($H = 6$h)
Fundamental Ceiling:
Even a 100% Perfect AI Oracle
Can Precede Treatment in <4% of
All Sepsis ICU Admissions
Theoretical Maximum Coverage $C_{max}(\delta, b)$ (%)
0
2
4
6
8
10
12
14
16
18
Required Actionable Buffer Ahead of Treatment (δ, Hours)
C. Temporal Gap & Accounting Decomposition
Treatment Simultaneous with Sepsis-3 ($G = 0$)
6h Gap Boundary ($G = 6$h)
Median Lead Gap ($G = +9.1$h)
Domain of Negative Lead Time ($L_{treat} < 0$)
Probability Density
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
−10
0
10
20
30
40
50
Diagnostic Lead Gap: $G = T_{sepsis3} - T_{treatment}$ [Hours]